Special Numbers

Year 4 · **Homework Worksheet**

Worked Example

Square, Cube & Triangular Numbers: Find the 6th triangular number, then state whether it is also a square number.

Answer: The 6th triangular number is 1+2+3+4+5+6 = 21. Since no whole number squared equals 21 (4^2=16, 5^2=25), 21 is not a square number.

Square & Cube Basics (15 questions)

1. Write down the first six cube numbers.
2. What is 9^2?
3. What is the square root of 64?
4. What is 3^3 (3 cubed)? A) 6 B) 9 C) 18 D) 27 E) 36
5. What is the cube root of 8?
6. Is 50 a square number? Explain how you know.
7. Write down the first five triangular numbers.
8. What is the 6th triangular number? A) 15 B) 18 C) 21 D) 24 E) 28
9. What is 12^2?
10. Which of these is a cube number? A) 27 B) 30 C) 40 D) 50 E) 60
11. Select ALL that apply: which of these numbers are square numbers? A) 4 B) 12 C) 36 D) 44 E) 49
12. What is the square root of 121?
13. What is 5^3 (5 cubed)?
14. Select ALL that apply: which of these numbers are triangular numbers? A) 3 B) 5 C) 10 D) 12 E) 21
15. What is $4^2 + 2^3$? A) 22 B) 23 C) 24 D) 25 E) 26

Patterns & Classifying (15 questions)

1. A number is both a square number and an odd number, and it lies between 20 and 40. What is the number?
2. The sum of the first 4 odd numbers is 1 + 3 + 5 + 7 = 16. Use this pattern to write the sum of the first 7 odd numbers without adding them one by one.
3. What is the 9th triangular number?
4. Which of these numbers is a triangular number? A) 26 B) 27 C) 28 D) 29 E) 30
5. Select ALL that apply: which of these statements about 100 are true? A) 100 is a square number. B) 100 is a cube number. C) The square root of 100 is 10. D) 99 is a cube number. E) 100 is an odd number.
6. Find two triangular numbers that add together to make 16.
7. What is the difference between the 5th and 4th square numbers?
8. A number is a cube number greater than 100 and less than 150. What is the number? A) 64 B) 81 C) 100 D) 121 E) 125

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9. The 7th triangular number is 28. What is the 8th triangular number?

10. Select ALL that apply: which of these numbers are both square numbers and triangular numbers? A) 1 B) 4 C) 9 D) 16 E) 36

11. What is the sum of the first 6 odd numbers? What type of special number does this give?

12. Which square number, when you find its square root, gives an even number greater than 8? A) 49 B) 64 C) 81 D) 100 E) 121

13. A display of cans is stacked in a triangular pattern: 1 can in the top row, 2 in the next, 3 in the next, and so on. How many cans are in a stack with 7 rows?

14. What is $6^2 - 4^2$? A) 18 B) 19 C) 20 D) 21 E) 22

15. Explain, using an example, why a triangular number plus the next triangular number always gives a square number.

Standard / Advanced (15 questions)

1. Counting upward, which square number is the first one past 120?

2. Looking at cube numbers in order, which is the last one before you pass 90?

3. Among the options below, which square number has 4 as its square root? A) 9 B) 16 C) 25 D) 36

4. Given that the 14th triangular number is 105, what must the 15th triangular number be — without adding up all 15 terms?

5. Select ALL that apply: a value of n is a triangular number and also an odd number, somewhere between 0 and 30. Which of A to E fit both rules? A) 3 B) 6 C) 10 D) 15 E) 21

6. The 6th triangular number is 21. Work out $6^2 + 6$, and check whether it equals double the 6th triangular number.

7. A square-shaped mat covers 81 cm^2 in total. How long is one of its sides?

8. Which of these square numbers ends in the digit 9? A) 16 B) 25 C) 36 D) 49

9. Given that the 4th triangular number is 10, work out the 5th and 6th triangular numbers, and describe the pattern you used to get them.

10. A whole number n, when cubed, gives 343. What is n? A) 5 B) 6 C) 7 D) 8

11. Select ALL that apply: which of these statements about square and cube numbers are true? A) 100 is a square number. B) 27 is a cube number. C) 1 is both a square number and a cube number. D) 50 is a square number. E) 125 is a cube number.

12. Crates are piled so each layer has one more crate than the layer above it, starting with a single crate on top. For a pile 7 layers deep, how many crates does that take in total?

13. What whole number, when squared, gives 169? A) 11 B) 12 C) 13 D) 14

14. A number is both a triangular number and a square number. It is greater than 36 but less than 100. Is there such a number in this range? Explain your answer.

15. Remembering that the first n odd numbers always add up to n^2, work out the total of the first 9 odd numbers using this shortcut.

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